ADVANCED EXCEL · LESSON 2 OF 12

Data Entry, Data Validation & Data Cleaning

Stop bad data at the door with data validation and drop-down lists, then clean what already exists with Flash Fill, Text to Columns, Find & Replace, Go To Special, Remove Duplicates, Paste Special and Excel Tables.

● Live class 2 ✦ Excel for Microsoft 365, 2026 Beginner ⏱ 90 min

By the end of this lesson you can

- Enter data quickly and consistently with AutoFill, Flash Fill and custom lists
- Build drop-down lists, number and date limits and custom-formula rules with Data Validation
- Create dependent (cascading) drop-downs
- Clean messy text, split and combine columns, and remove duplicates safely
- Use Paste Special and Excel Tables as everyday cleaning tools

Datasets used

`employees_small.csv`

Garbage in, garbage out

Most time in real analytics is not spent on fancy charts; it is spent fixing data: extra spaces, inconsistent spellings (`Bangalore` , `Bengaluru` , `bengaluru`), numbers stored as text, duplicates and blanks. Surveys of data teams regularly report that cleaning and preparing data takes well over half of an analyst's time.

There are two ways to win this fight:



Prevent problems at entry time, and clean whatever slips through. Both are needed.

Fast, consistent data entry

AutoFill and the fill handle

Select a cell and drag the small square at its bottom-right corner (the **fill handle**). Excel continues the pattern it recognises.

You type	Drag gives you
`Jan`	Feb, Mar, Apr ...
`Monday`	Tuesday, Wednesday ...
`1` , `2` (select both)	3, 4, 5 ...
`Q1`	Q2, Q3, Q4, Q1 ...
`01-04-2026`	02-04-2026, 03-04-2026 ... (right-drag for months or years)
`Item 1`	Item 2, Item 3 ...
`=B2*12`	`=B3*12` , `=B4*12` ...

Double-click the fill handle to fill a formula down as far as the column next to it has data. This is the fastest way to copy a formula to 10,000 rows.

TIP **Custom lists**

File → Options → Advanced → **Edit Custom Lists** lets you add your own sequences, for example `North, South, East, West` or your company's department order. AutoFill and custom sort then follow that order.

Flash Fill (Ctrl + E)

Flash Fill learns a pattern from one or two examples you type. It is perfect for splitting names, extracting codes or reformatting text.

Flash Fill: type the first answer in each column, then press Ctrl + E

	A	B	C	D	E
1	**Full name**	**Email**	**First name**	**Last name**	**Username**
2	Kavya Iyer	kavya.iyer@abc.in	Kavya	Iyer	kiyer
3	Imran Reddy	imran.reddy@abc.in	Imran	Reddy	ireddy
4	Harpreet Kulkarni	harpreet.k@abc.in	Harpreet	Kulkarni	hkulkarni
5	Diya Iyer	diya.iyer@abc.in	Diya	Iyer	diyer

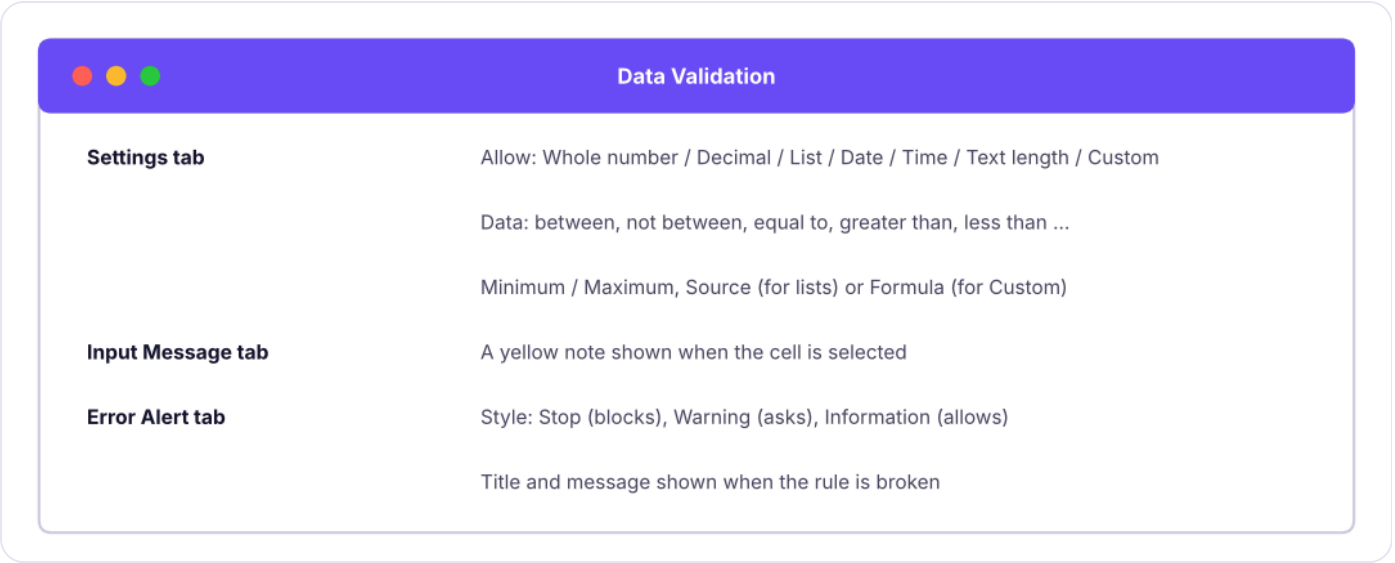
Purple cells are typed by you; green cells are filled by Flash Fill.

WATCH OUT **Flash Fill is not a formula**

Flash Fill writes **static values**. If the source changes, the results do not update. Use formulas (Lesson 5) or Power Query (Lesson 12) when the data will be refreshed.

Data Validation: rules for what can be typed

Data → Data Validation (shortcut `Alt, A, V, V`) attaches a rule to cells. Anyone who types something that breaks the rule is stopped or warned.



The three tabs of the Data Validation dialog.

The validation types you will use most

Allow	Example rule	Business use
Whole number	between 18 and 65	Age of an employee
Decimal	between 0 and 0.5	Discount rate
List	`Sales,Finance,IT,HR,Operations`	Department drop-down
Date	between `01-04-2026` and `31-03-2027`	Entries inside the financial year
Time	between 09:00 and 18:00	Office hours
Text length	equal to 6	PIN code, employee code
Custom	`=COUNTIF($A$2:$A$500,A2)=1`	No duplicate IDs

Building a drop-down list, step by step

STEP BY STEP **Department drop-down**

1. On a sheet called `Lists`, type the allowed values in a column (`Sales`, `Finance`, `IT`, `HR`, `Operations`) and convert it to a Table (`Ctrl + T`) named `tblDept`. A Table grows automatically when you add a department.
2. Select the cells where people will enter a department (for example `C2:C500`).
3. Data → Data Validation → Allow: **List**.
4. Source: `=INDIRECT("tblDept[Department]")` or simply select the range `=Lists!$A$2:$A$6`.
5. Keep **In-cell dropdown** ticked. Add an Input Message ("Choose a department") and an Error Alert (Stop, "Please pick a department from the list").

Data entry sheet with validation rules (rules shown in row 2)

	A	B	C	D	E	F	G
1	**Emp ID**	**Name**	**Department**	**Age**	**Join date**	**Discount**	**PIN code**
2	*Text length = 4*	*Any text*	*List: tblDept*	*Whole 18–65*	*Date ≥ 01-01-2000*	*Decimal 0–0.5*	*Text length =*
3	E101	Kavya Iyer	HR	29	28-02-2016	0.1	40000
4	E102	Imran Reddy	Fin	17	14-12-2019	0.75	1100

Row 3 passes every rule (green). In row 4, `Fin` is not in the list, 17 is under the minimum age, 0.75 is above the maximum discount and `11002` has only 5 characters, so Excel would block each of these entries (red).

Custom formula rules

The **Custom** option accepts any formula that returns TRUE (allow) or FALSE (block). Write the formula for the **first** selected cell; Excel adjusts it for the others.

Rule	Custom formula (cells A2:A500 selected)
No duplicates	`=COUNTIF($A$2:$A$500,A2)=1`
Must start with "E" and be 4 characters	`=AND(LEFT(A2,1)="E",LEN(A2)=4)`
Numbers only (for a text-formatted column)	`=ISNUMBER(--A2)`
No spaces	`=ISERROR(FIND(" ",A2))`
Weekdays only (dates)	`=WEEKDAY(A2,2)<6`
Total of a budget column cannot exceed 1,00,000	`=SUM($B$2:$B$50)⩽100000`
Email must contain "@" and "."	`=AND(ISNUMBER(FIND("@",A2)),ISNUMBER(FIND(".",A2)))`

Dependent (cascading) drop-downs

A second list that changes with the first: choose a **Category**, and the **Sub-category** list shows only matching items.

Lists sheet: each column is a named range (Formulas → Create from Selection → Top row)

	A	B	C
1	**Furniture**	**Office_Supplies**	**Technology**
2	Chairs	Binders	Phones
3	Tables	Paper	Accessories
4	Bookcases	Storage	Machines
5	Furnishings	Art	Copiers
6		Appliances	

STEP BY STEP **Make it work**

1. Select the lists (with headers) → Formulas → **Create from Selection** → Top row. Excel creates the names `Furniture`, `Office_Supplies` and `Technology`.
2. Category cell A2: Data Validation → List → Source: `Furniture,Office_Supplies,Technology`.
3. Sub-category cell B2: Data Validation → List → Source: `=INDIRECT(A2)`.
4. `INDIRECT` turns the text in A2 into a reference to the named range with that name.

2026 UPDATE **Dynamic-array version (Microsoft 365)**

With a two-column mapping table (`Category`, `SubCategory`), put `=FILTER(Map[SubCategory],Map[Category]=A2)` in a helper cell and point the validation source to it with the spill operator, for example `=$H$2#`. No named ranges are needed, and new items appear automatically.

Finding data that already breaks the rules

Validation only checks **new** typing. Data pasted in, or entered before the rule existed, is not checked. Use **Data → Data Validation → Circle Invalid Data** to draw red circles around existing violations.

COMMON MISTAKES **Validation is not security**

Pasting over a validated cell replaces the rule. Use Paste Special → Values when pasting into validated areas, and protect the sheet (Lesson 11) if people must not change rules.

Cleaning text with formulas

These functions fix most text problems. Each formula below is written for row 2 and copied down.

Common text-cleaning formulas

	A	B	C	D	E
1	Raw value	=TRIM(A2)	=PROPER(TRIM(A2))	=SUBSTITUTE(A2,"-","")	=UPPER(A2)
2	bengaluru	bengaluru	Bengaluru	bengaluru	BENGALURU
3	NEW DELHI	NEW DELHI	New Delhi	NEW DELHI	NEW DELHI
4	pune-411001	pune-411001	Pune-411001	pune411001	PUNE-411001
5	mUMBAI	mUMBAI	Mumbai	mUMBAI	MUMBAI

```
B2   =TRIM(A2)
C2   =PROPER(TRIM(A2))
D2   =SUBSTITUTE(A2,"-","")
E2   =UPPER(A2)
```

Each pattern is copied down/across to the other cells in its column.

- `TRIM` removes spaces at the start and end and squeezes repeated inner spaces to one.
- `PROPER` capitalises the first letter of each word; `UPPER` and `LOWER` change the case of everything.
- `SUBSTITUTE(text, old, new)` replaces every occurrence of `old`.

- `CLEAN` removes non-printing characters such as line breaks from copied web data.
- The non-breaking space (character 160) from websites is not removed by TRIM. Use `=TRIM(SUBSTITUTE(A2,CHAR(160)," "))`.

Text to Columns

Data → Text to Columns splits one column into several, either by a **delimiter** (comma, space, `-`, `|`) or by **fixed width**.

Before: one column with a delimiter

	A	B	C
1	**Order info**		
2	IN-2025-00042	Pune	Technology
3	IN-2025-00043	Delhi	Furniture

STEP BY STEP **Split by a delimiter**

1. Select the column. Data → Text to Columns.
2. Choose **Delimited** → Next → tick **Other** and type `|` (or comma, tab, space).
3. In step 3, set each column's type: keep IDs and codes as **Text** to protect leading zeros; set dates to **Date: DMY**.
4. Choose the destination cell (so the original is not overwritten) → Finish.

Text to Columns is also a quick fix for numbers stored as text: select the column, Text to Columns, and click **Finish** immediately. Excel re-reads every value as a number.

2026 UPDATE **TEXTSPLIT, TEXTBEFORE, TEXTAFTER (Microsoft 365)**

Formula versions that update automatically: `=TEXTSPLIT(A2,"|")` spills the parts across columns; `=TEXTBEFORE(A2,"-")` and `=TEXTAFTER(A2,"-",-1)` return the text before the first or after the last hyphen. See Lesson 5.

Find & Replace and wildcards

`Ctrl + H` replaces text in the selection (or the whole sheet). Click **Options** to match case, match the entire cell, or search in formulas versus values.

Wildcard	Meaning	Example
*	Any number of characters	Bang* finds Bangalore, Bangaluru
?	Exactly one character	E10? finds E101 to E109
~	Treat the next character literally	~* finds a real asterisk

Example: replace Bang* with Bengaluru (with **Match entire cell contents** ticked) to standardise every spelling in one step.

Go To Special: select blanks, constants, formulas, errors

F5 → Special (or Ctrl + G → Special) selects cells by type. The most useful trick in data cleaning is **filling blanks with the value above**:

Before: category written only once per block

	A	B	C
1	**Category**	**Product**	**Sales**
2	Furniture	Chair A	12000
3		Table B	18500
4		Bookcase C	9400
5	Technology	Phone X	25200
6		Router Y	4300

STEP BY STEP **Fill blanks with the value above**

1. Select A2:A6. Press F5 → Special → Blanks → OK. Only the empty cells are selected (yellow).
2. Type = and press the **Up arrow** (the formula becomes =A2).
3. Press Ctrl + Enter to put the formula in every selected blank at once.
4. Select the column, copy, and Paste Special → **Values** to freeze the result.

Other Go To Special options: **Constants** (typed values only, great for clearing inputs but keeping formulas), **Formulas** (to check or protect them), **Errors**, **Visible cells only** (copy a filtered list without hidden rows; shortcut Alt + ;), **Row differences**.

Removing duplicates

Data → Remove Duplicates deletes rows that repeat in the columns you choose and keeps the **first** occurrence.

Duplicate check before removing

	A	B	C	D
1	**Emp ID**	**Name**	**City**	**Is duplicate?**
2	E101	Kavya Iyer	Mumbai	First
3	E102	Imran Reddy	Delhi	First
4	E101	Kavya Iyer	Mumbai	Duplicate
5	E103	Harpreet Kulkarni	Mumbai	First
6	E102	Imran Reddy	Delhi	Duplicate

```
D2   =IF(COUNTIF($A$2:A2,A2)>1,"Duplicate","First")
D3   =IF(COUNTIF($A$2:A3,A3)>1,"Duplicate","First")
D4   =IF(COUNTIF($A$2:A4,A4)>1,"Duplicate","First")
D5   =IF(COUNTIF($A$2:A5,A5)>1,"Duplicate","First")
D6   =IF(COUNTIF($A$2:A6,A6)>1,"Duplicate","First")
```

The expanding range `$A$2:A2` counts how many times the ID has appeared **so far**, so the first appearance says "First" and later ones say "Duplicate". This lets you review duplicates before deleting anything.

> **WATCH OUT** **Always work on a copy**
>
> Remove Duplicates permanently deletes rows. Copy the sheet first, or use the formula above, conditional formatting (Home → Conditional Formatting → Highlight Duplicate Values) or `=UNIQUE()` (Lesson 6) to inspect duplicates without losing data.

Paste Special: the cleaning Swiss-army knife

`Ctrl + Alt + V` opens Paste Special.

Option	What it does	Typical use
Values (V)	Pastes results, not formulas	Freeze formula results; break links to other files
Formats (T)	Copies only formatting	Apply a header style to another table
Transpose (E)	Rows become columns	Turn a wide header row into a list
Multiply / Divide / Add / Subtract	Combines the copied number with each target cell	Convert rupees to lakhs: copy `100000`, select the amounts, Paste Special → Divide
Skip blanks	Empty copied cells do not overwrite targets	Merge two partially filled columns
Paste Link	Creates `=Sheet1!A1` references	Live link to a source cell

Excel Tables: structure that maintains itself

Convert any clean range with `Ctrl + T`. Tables are the best habit you can build in Excel:

- **Auto-expand**: new rows and columns join the table, so formulas, charts, pivot tables and validation lists include them automatically.
- **Structured references**: formulas read like English, for example `=SUM(tblSales[Amount])` or `=[@Price]*[@Qty]`.
- **Calculated columns**: type a formula once and it fills the whole column.
- **Total Row**: Table Design → Total Row adds a row with drop-down summaries that respect filters (they use `SUBTOTAL`).
- **Banded rows, filter buttons and a frozen header** come for free.

An Excel Table named tblOrders with a calculated column and Total Row

	A	B	C	D	E
1	OrderID	Product	Price	Qty	Amount
2	1001	Chair	3500	4	14,000
3	1002	Desk	12500	1	12,500
4	1003	Lamp	1200	6	7,200
5	1004	Shelf	5400	2	10,800
6	**Total**			13	44,500

```
E2   =C2*D2
E3   =C3*D3
E4   =C4*D4
E5   =C5*D5
D6   =SUBTOTAL(109,D2:D5)
```

```
E6   =SUBTOTAL(109,E2:E5)
```

In a real Table, the Amount formula appears as `=[@Price]*[@Qty]`. The Total Row uses `SUBTOTAL(109, …)`, which sums only visible rows, so totals change when you filter.

DO IT WITH AI

Do it with AI

Copilot in Excel or any chat assistant can draft cleaning steps. A good prompt: *"I have a column of Indian city names with inconsistent spelling, extra spaces and mixed case (examples: ' bengaluru', 'BANGALORE', 'Bengaluru '). Give me an Excel formula to standardise them to proper case, and a list of Find & Replace steps to map old names to new ones."* Always test the answer on a copy of your data.

Summary

KEY TAKEAWAYS

- **Prevent** bad data with Data Validation (lists, limits, custom formulas) and **cure** the rest with cleaning tools.
- Validation lists should point to a Table so they grow automatically; use `INDIRECT` or `FILTER` for dependent drop-downs.
- Validation does not check pasted or old data: use **Circle Invalid Data**.
- `TRIM`, `CLEAN`, `PROPER`, `SUBSTITUTE` fix most text; Text to Columns and Flash Fill split data.
- **Go To Special → Blanks** then =↑ and `Ctrl + Enter` fills blank cells with the value above.
- Inspect duplicates with `COUNTIF` before using Remove Duplicates; always keep a copy.
- Turn every dataset into an **Excel Table** (`Ctrl + T`).

Quick quiz

6 questions · tap an answer to check it

1. Which validation setting stops users entering a duplicate employee ID in A2:A500?

A List with source A2:A500

B Custom: `=COUNTIF($A$2:$A$500,A2)=1`

C Text length equal to 4

D Whole number between 1 and 500

Answer: B. COUNTIF counts how many times the value appears; allowing only a count of 1 blocks duplicates.

2. You pasted 2,000 rows into a validated column. How do you find values that break the rule?

A Re-create the validation rule

B Data Validation → Circle Invalid Data

C Press F9

D Use Remove Duplicates

Answer: B. Validation only checks typing. Circle Invalid Data marks existing violations.

3. What does `=TRIM(" New Delhi ")` return?

A "New Delhi"

B "New Delhi"

C "NewDelhi"

D " New Delhi "

Answer: B. TRIM removes leading and trailing spaces and reduces repeated inner spaces to one.

4. Which feature fills blank cells in a column with the value from the cell above in a few keystrokes?

A Flash Fill

B Remove Duplicates

C Go To Special → Blanks, then = + Up arrow + Ctrl + Enter

D Text to Columns

Answer: C. Go To Special selects only the blanks; Ctrl + Enter writes the relative formula into all of them.

5. What is the main downside of Flash Fill?

A It only works on numbers

B Results are static values that do not update when the source changes

C It needs a macro

D It cannot split text

Answer: B. Use formulas or Power Query for refreshable cleaning.

6. Why should a drop-down list's source be an Excel Table?

A The list grows automatically when new items are added

B Tables are required for validation

C It makes the drop-down faster

D It prevents duplicates

Answer: A. Tables expand automatically, so new departments appear in the drop-down without editing the rule.

EXERCISE

Validation set-up

Using `employees_small.csv`, add validation so that: department must be one of Sales, Finance, IT, HR, Operations; rating must be a whole number from 1 to 5; join date cannot be in the future; and employee IDs are unique.

Show solution (worked answer)

- Department (C2:C21): List, source `Sales,Finance,IT,HR,Operations`.
- Rating (G2:G21): Whole number, between 1 and 5.
- Join date (E2:E21): Date, less than or equal to `=TODAY()`.
- Emp ID (A2:A21): Custom, `=COUNTIF($A$2:$A$21,A2)=1`. Add an Error Alert (Stop) for each with a helpful message, then run Circle Invalid Data to check the existing rows.

EXERCISE

Clean the city column

A column contains: `mumbai`, `MUMBAI`, `Bombay`, `Mumbai`, `mumbai city`. Produce one clean value, `Mumbai`, for all five.

Show solution (worked answer)

1. Helper formula `=PROPER(TRIM(A2))` gives `Mumbai`, `Mumbai`, `Bombay`, `Mumbai`, `Mumbai City`.
2. Paste Special → Values over the original.
3. Find & Replace with Match entire cell contents: `Bombay` → `Mumbai`, `Mumbai City` → `Mumbai`. For data that refreshes every week, build the same steps in Power Query so they repeat automatically.

INTERVIEW CORNER

Questions you may be asked

Q: How would you stop users from entering invalid data in a shared tracker? Use Data Validation (drop-down lists from a Table, number and date limits, custom formulas for uniqueness), add input messages, set error alerts to Stop, and protect the sheet so the rules cannot be removed. Check older data with Circle Invalid Data.

Q: How do you remove duplicates without losing information? First flag them with `COUNTIF` or conditional formatting and review which record to keep; decide the key columns that define a duplicate; work on a copy; then use Remove Duplicates (or `UNIQUE`) on those columns.

Q: What is the difference between TRIM and CLEAN? TRIM removes extra spaces; CLEAN removes non-printing characters such as line breaks. Web data often needs both, plus `SUBSTITUTE(…,CHAR(160)," ")` for non-breaking spaces.

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